

D7056E Assignment 2: Replication Study

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1 Model Performance

I used a sample of 20,000 reviews from the given dataset. The final dataset was processed using Random Forest for prediction and LDA for topic extraction after reviews with at least two votes were filtered out, which improves the quality of the data.

Data Process: I started with 20,000 raw reviews. After filtering for quality (reviews with at least 2 votes), the dataset was reduced to 8,811 rows.

```
Total reviews: 20000
Reviews after filtering: 8811
```

Figure 1: Total reviews before and after filtering

Accuracy: The accuracy of my model is $\sim 68\%$.

```
accuracy          0.68      1763
```

Figure 2: Model's Accuracy

Performance: Class 1 (Helpful): With 0.85 recall and precision of 0.70
Class 0 (Not Helpful): With 0.36 recall and precision of 0.58.

```
Performance:
           precision    recall  f1-score   support

0         0.58         0.36         0.44         634
1         0.70         0.85         0.77        1129
```

Figure 3: Model's Performance

2 Comparison with Original Study

Original Results: The predictive performance (AUC) of the original study was roughly 73%.

Discrepancy Analysis: My model performed slightly lower (68%) than the original study (73%), resulting in a difference of $\sim 5\%$. The reasons are likely:

- **Sample Size:** The original study used 495,358 reviews, whereas I used a subset of 20,000 for computational efficiency.

- **Number of Topics:** I used 10 topics to keep the model interpretable, while the original study utilized 100 topics.

3 Reflection

3.1 3 Things I Learned

I learned that:

- **Data Quality:** The filtering part is essential; removing reviews with fewer than two votes greatly increases the training labels' reliability. Because one review may be a coincidence, but two or more reviews give a real indication.
- **Text to Math:** How to use Bag of Words to convert unprocessed human text into numerical vectors for machine processing.
- **Lemmatization:** Converting words like 'running' to their root 'run' reduces "noise" and helps the model group similar meanings.

3.2 Questions I Still Have

Here are two questions that I still have:

- Given the 68% accuracy, how reliable is this model if applied to a completely different category of reviews instead of Video Games?
- Is Random Forest the best model here, or would an algorithm handle the interaction between 'Topic' and 'Overall rating' more effectively?

3.3 Thing I Enjoyed

Here is 1 thing I enjoyed:

I liked the text cleaning step because it made the most crucial words stand out by eliminating boring words like "the" and "is," etc.

4 Use of AI Tools

For this assignment, I utilized Gemini (AI) and ChatGPT in the following ways:

- **Code Structure:** To help structure the Python pipeline, specifically for correctly implementing the gensim LDA model and the scikit-learn Random Forest.
- **Concept Clarification:** Explain the logic behind doc2bow and the `max` function used to assign the dominant topic to each review.
- **Formatting:** Organizing the final report into a clear, readable format.